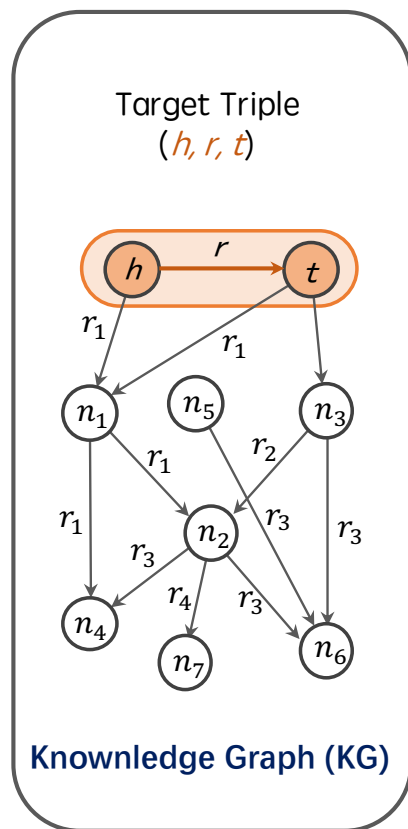


Phase 1: Knowledge Graph Input



Phase 2: Global Structure Prior Modeling

1) Structural Role Calculate

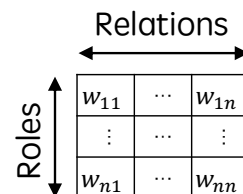
Degree

PageRank

Clustering Coefficient

Relation Diversity

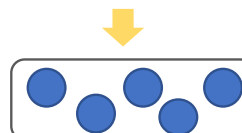
2) Relation-Role TF-IDF Modeling



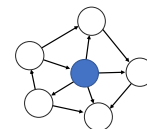
3) Diffusion Source Selection

↑ Importance

↓ Consistency



4) Global Structure Diffusion

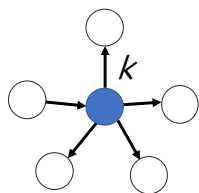


Propagation under TF-IDF Constraints

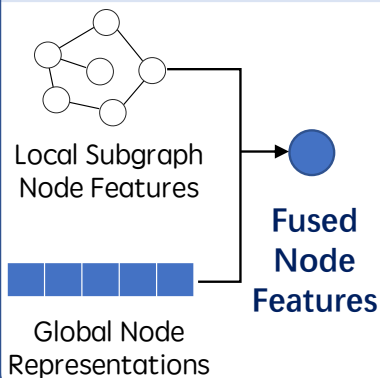


Phase 3: Structure-Guided Subgraph Completion

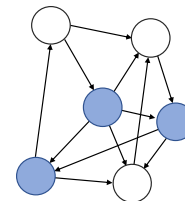
1) k-hop Subgraph Extraction



2) Relation-Role TF-IDF Modeling



3) GNN Inference



4) Link Prediction Output

Score (h, r, t)

Prediction Result

Main Legend:



Module: Process Step;



Specific Connection/Relationship



Target Element